A DIRECTED COLIMIT OF ALGEBRAIC SPACES GLUED BY THICKENINGS IS A FORMAL ALGEBRAIC SPACE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof contains the author's parenthetical '(details omitted)' attached to the claim that the morphism from the colimit of the pieces to the colimit of the members is etale; it is reproduced verbatim and recorded as an obligation of the card.

Editorial note. The proof displays one cartesian diagram; it lies inside the counted span and is reproduced verbatim.

Editorial note. The counted spans carry 4 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Algebraic spaces can be covered by etale morphisms from affine schemes. Properties of Spaces, Lemma spaces-properties-lemma-cover-by-union-affines, invoked at proof lines 2984-2987. Supplies the starting covering of a member of the system.
- Topological invariance of the etale site under a thickening. More on Morphisms of Spaces, Theorem spaces-more-morphisms-theorem-topological-invariance, invoked at proof lines 2991-2997. Propagates the etale covering along the system and gives uniqueness of the propagated diagrams.
- The fibre product of an affine scheme along a thickening is affine. Limits of Spaces, Proposition spaces-limits-proposition-affine, invoked at proof lines 3000-3003. Shows each piece of the propagated covering is an affine scheme.

Source reference. Original proposition 07VT: sources/stacks-chow/tags.txt, source lines 10018–10018. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original theorem 05ZH: sources/stacks-chow/tags.txt, source lines 7691–7691. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:formal-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 03FX: sources/stacks-chow/tags.txt, source lines 4508–4508. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let S be a scheme. Suppose given a directed set Λ and a system of algebraic spaces $(X_\lambda, f_{\lambda\mu})$ over Λ where each $f_{\lambda\mu} : X_\lambda \to X_\mu$ is a thickening. Then $X = \operatorname{colim}_{\lambda\in\Lambda} X_\lambda$ is a formal algebraic space over S.*

Proof. Since we take the colimit in the category of fppf sheaves, we see that X is a sheaf. Choose and fix $\lambda \in \Lambda$. Choose an étale covering $\{X_{i,\lambda} \to X_\lambda\}$ where X_i is an affine scheme over S, see Properties of Spaces, Lemma 03FX. For each $\mu \geq \lambda$ there exists a cartesian diagram

$$\begin{array}{ccc} X_{i,\lambda} & \longrightarrow & X_{i,\mu} \\ \downarrow & & \downarrow \\ X_\lambda & \longrightarrow & X_\mu \end{array}$$

with étale vertical arrows, see More on Morphisms of Spaces, Theorem 05ZH (this also uses that a thickening is a surjective closed immersion which satisfies the conditions of the theorem). Moreover, these diagrams are unique up to unique isomorphism and hence $X_{i,\mu} = X_\mu \times_{X_{\mu'}} X_{i,\mu'}$ for $\mu' \geq \mu$. The morphisms $X_{i,\mu} \to X_{i,\mu'}$ is a thickening as a base change of a thickening. Each $X_{i,\mu}$ is an affine scheme by Limits of Spaces, Proposition 07VT and the fact that $X_{i,\lambda}$ is affine. Set $X_i = \operatorname{colim}_{\mu\geq\lambda} X_{i,\mu}$. Then X_i is an affine formal algebraic space. The morphism $X_i \to X$ is étale because given an affine scheme U any $U \to X$ factors through X_μ for some $\mu \geq \lambda$ (details omitted). In this way we see that X is a formal algebraic space. □